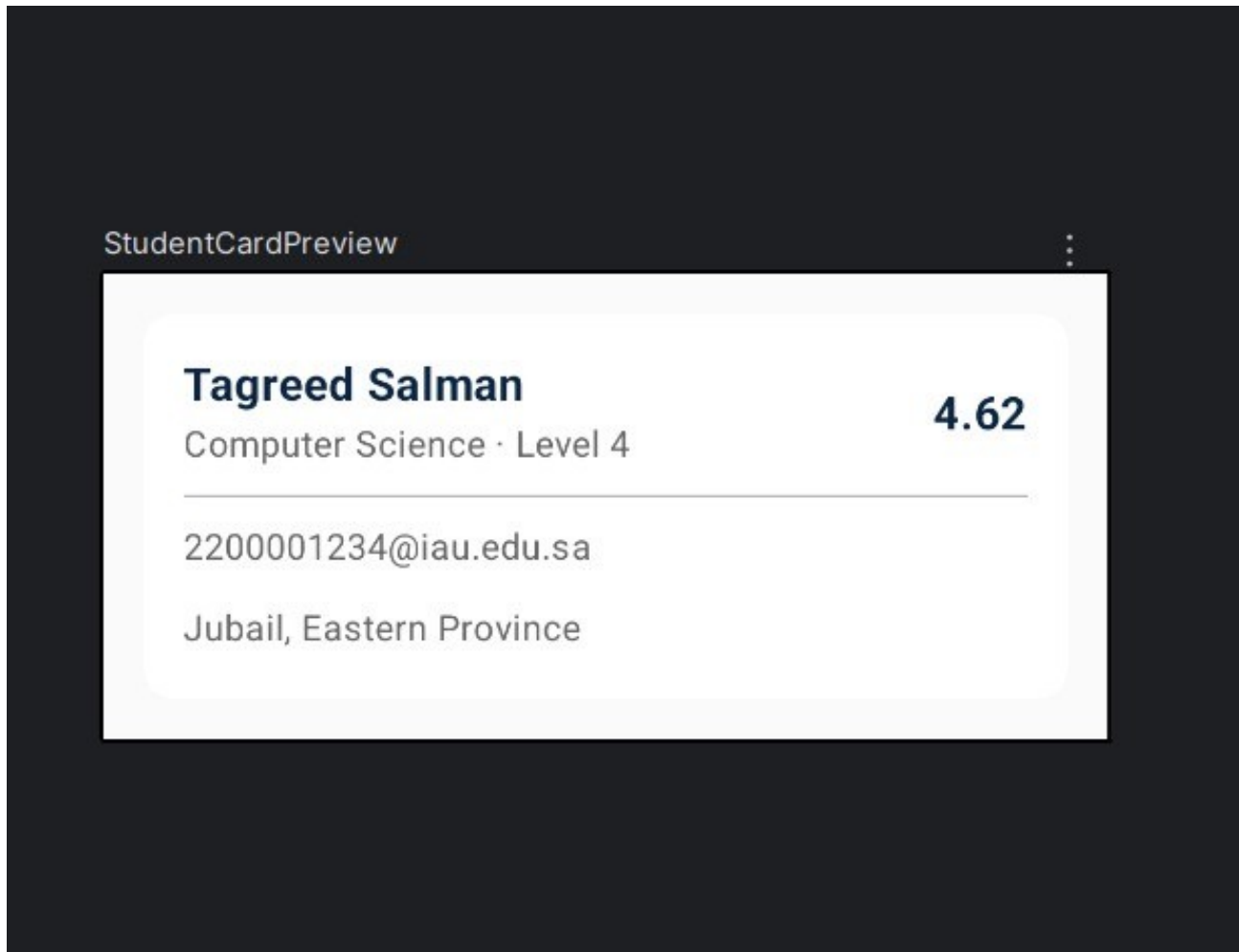


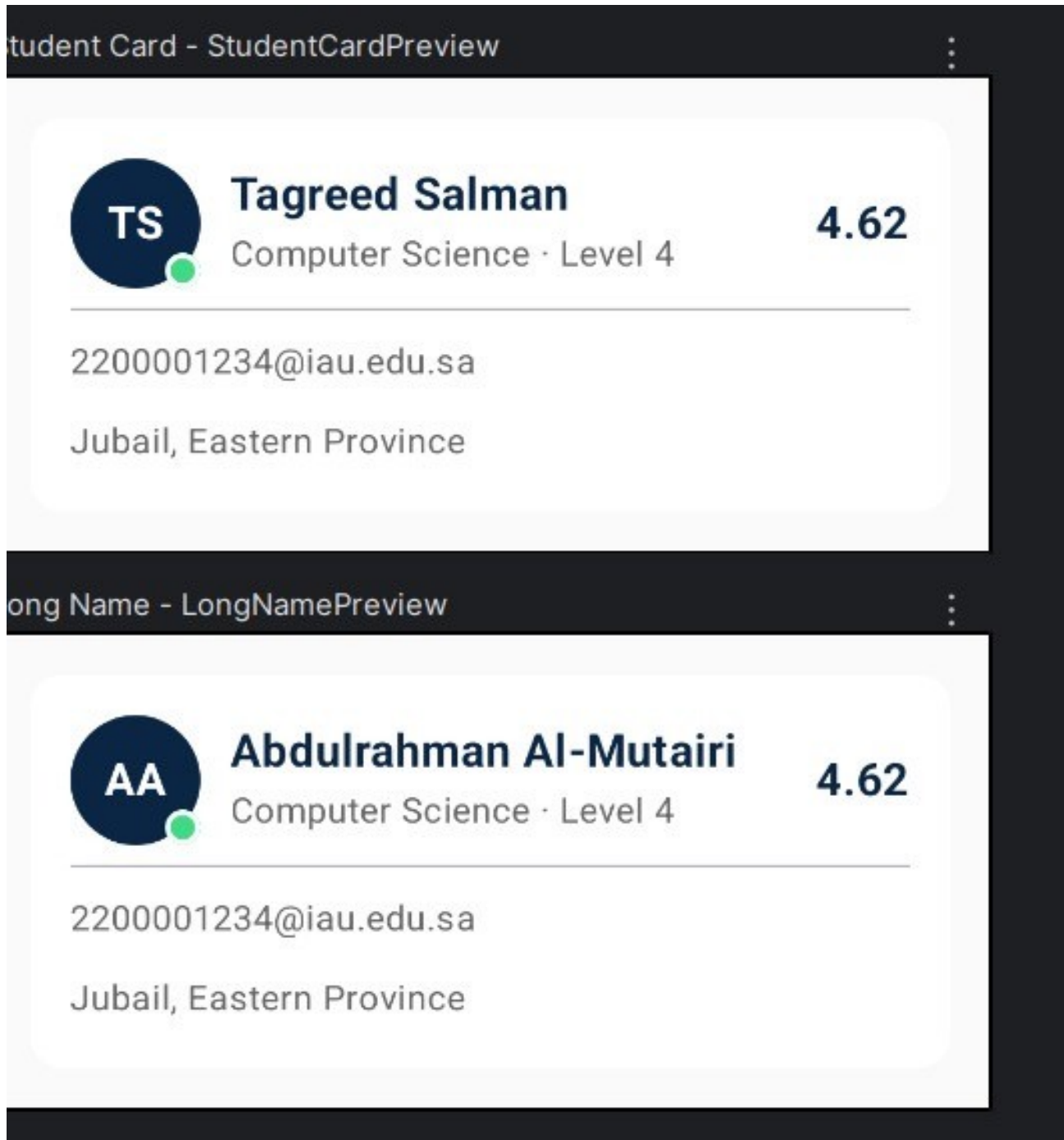
Jetpack Compose Lab Report

Student cards, modifier order, Material Design styling and profile screens.

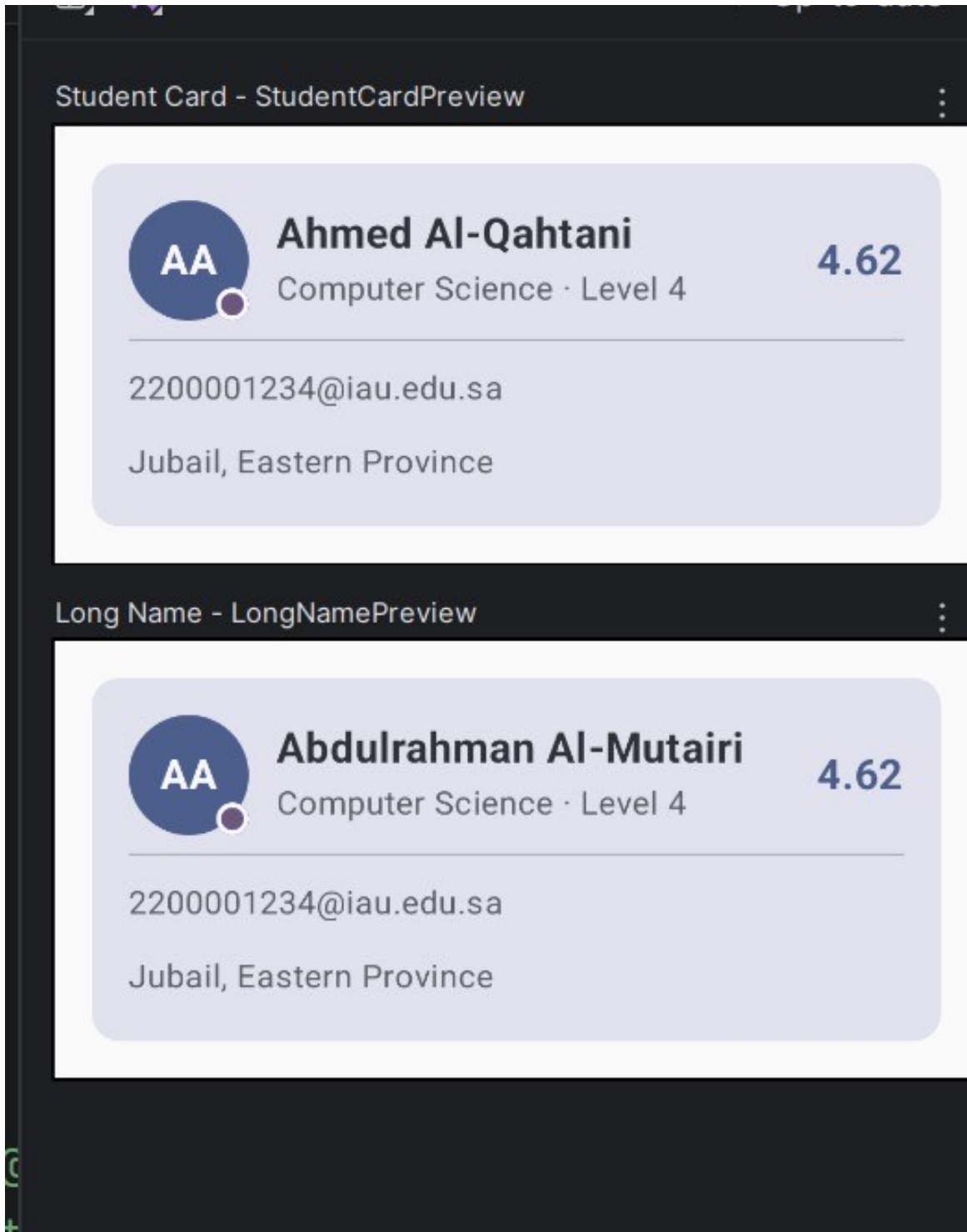
Student Card



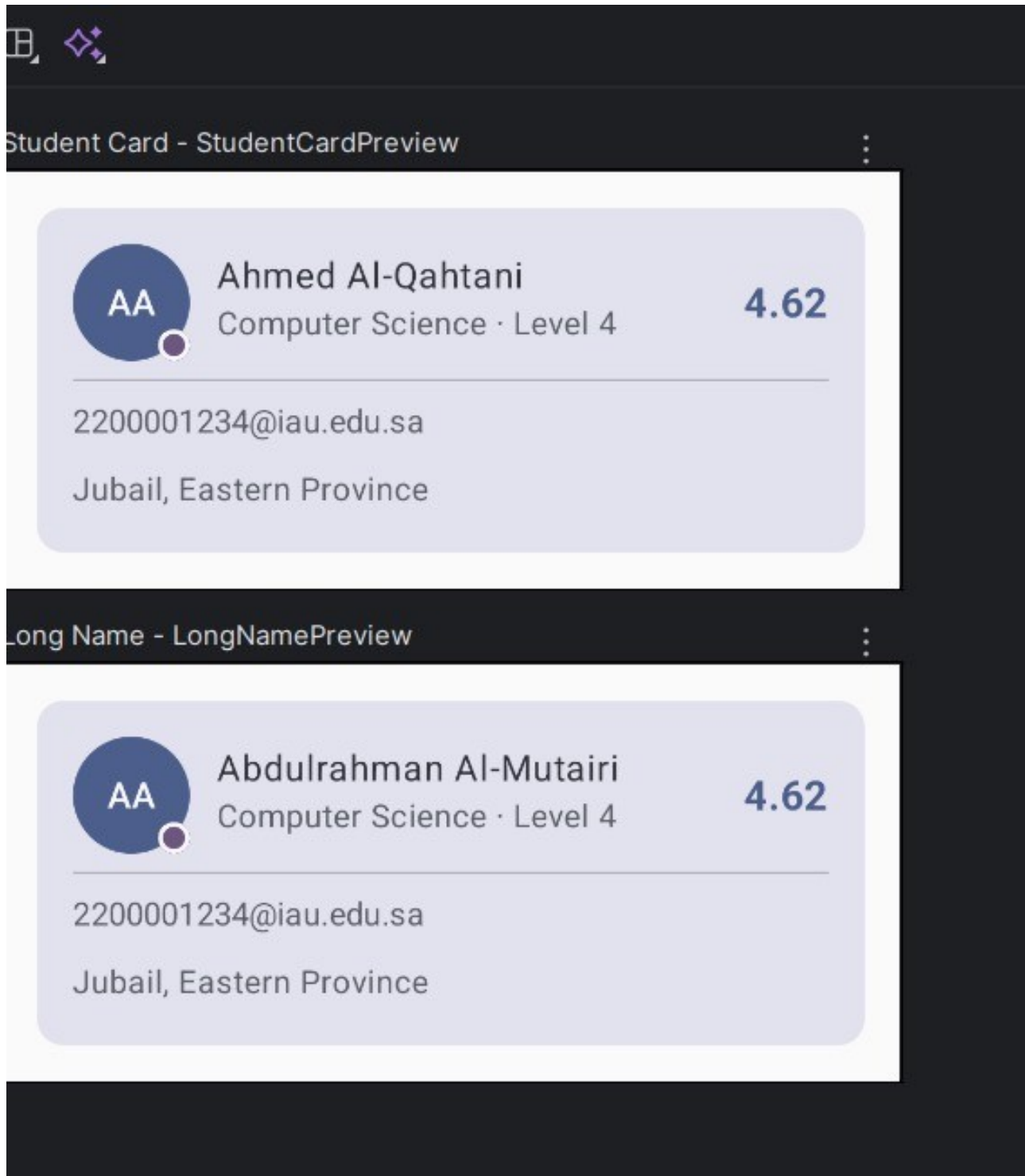
Avatar and Status Badge



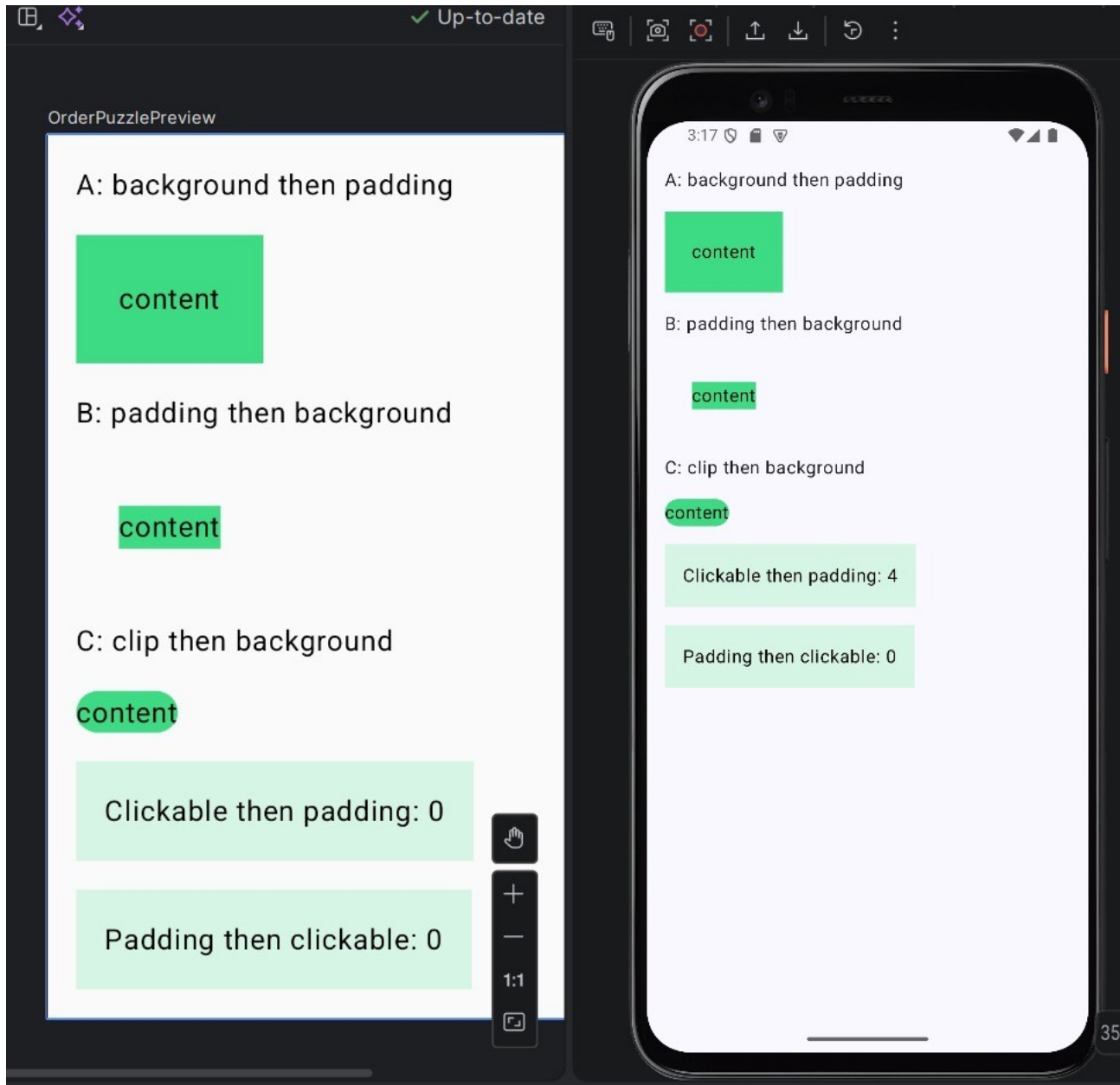
Material Theme Colours



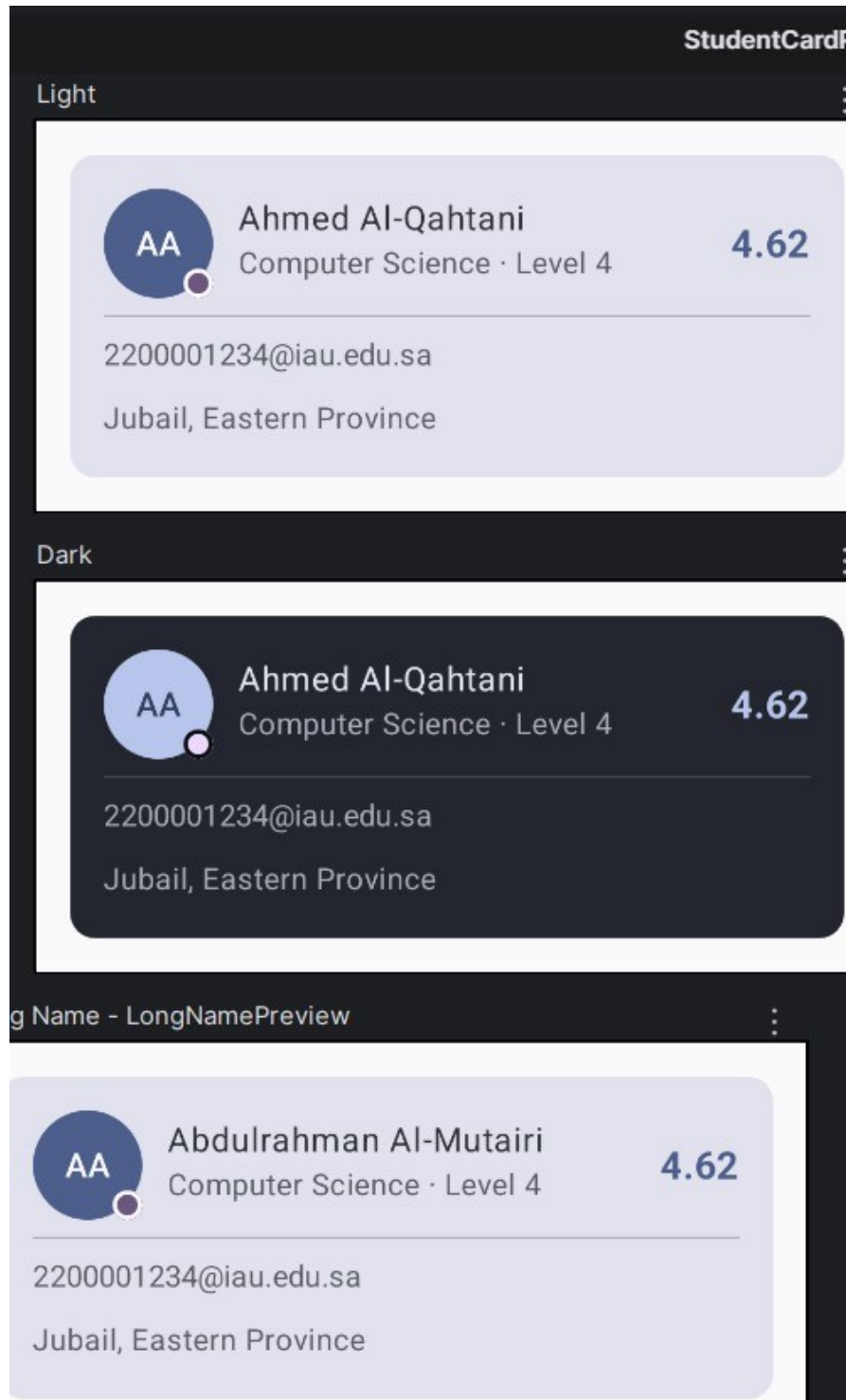
Material Theme Typography



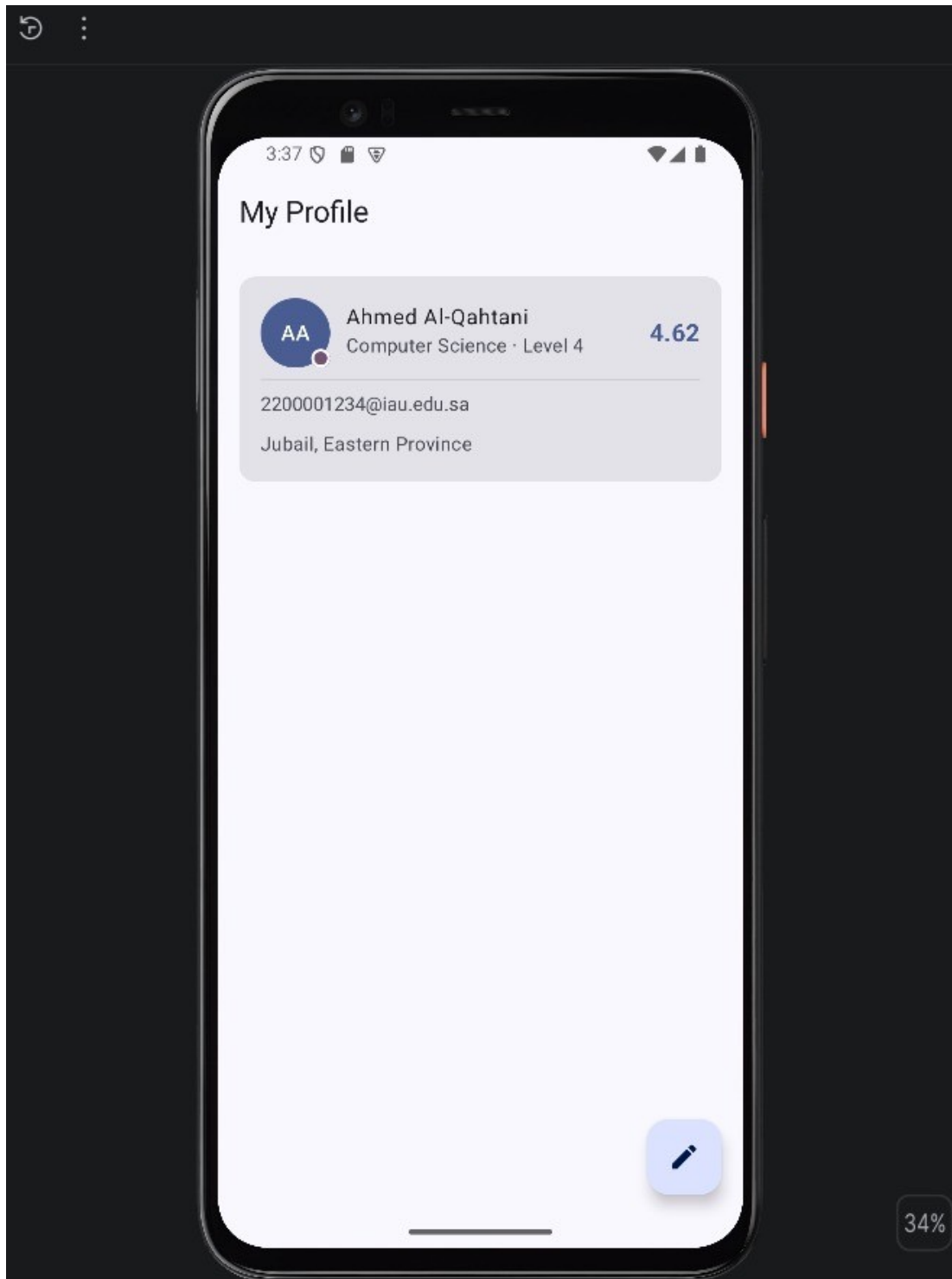
Modifier Order and Tap Targets



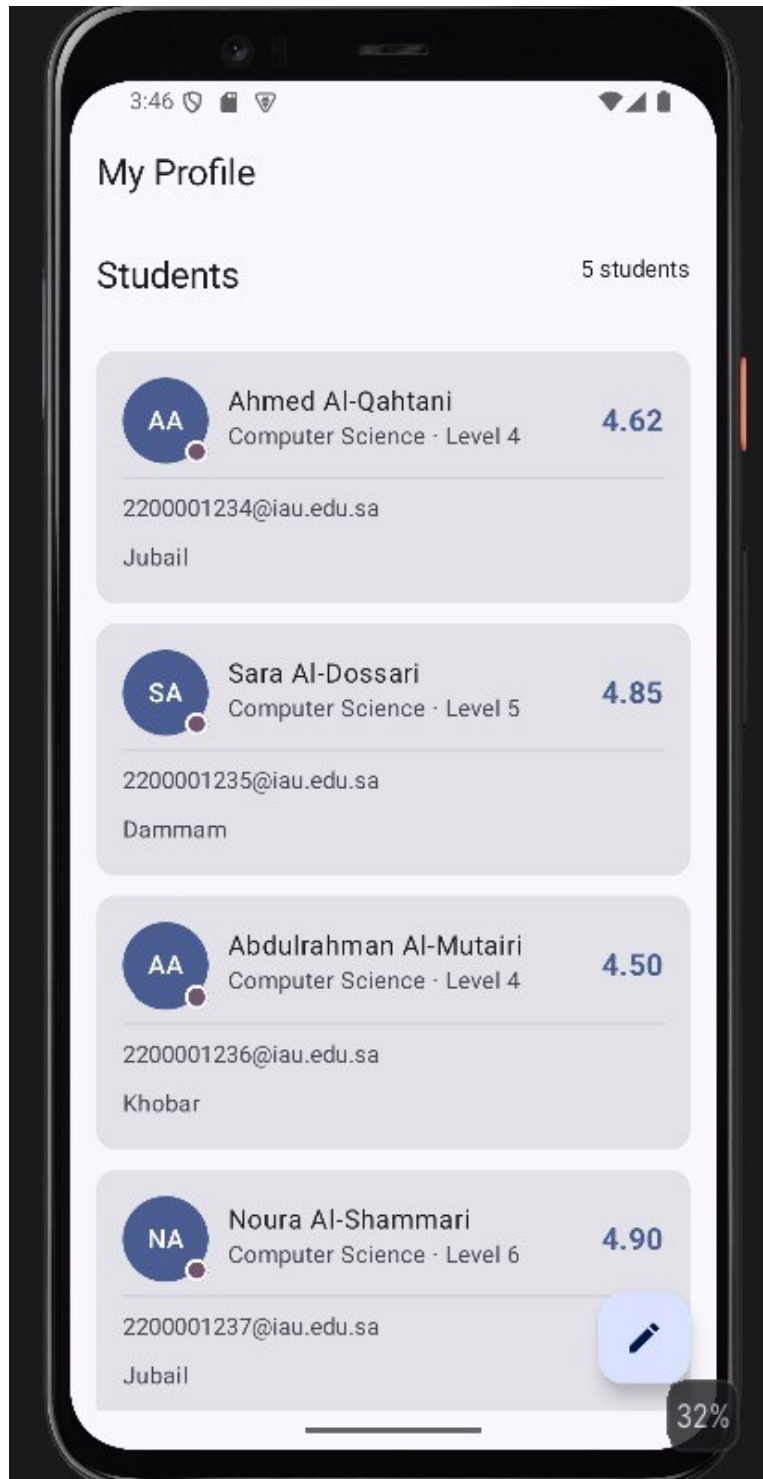
Light and Dark Card Previews



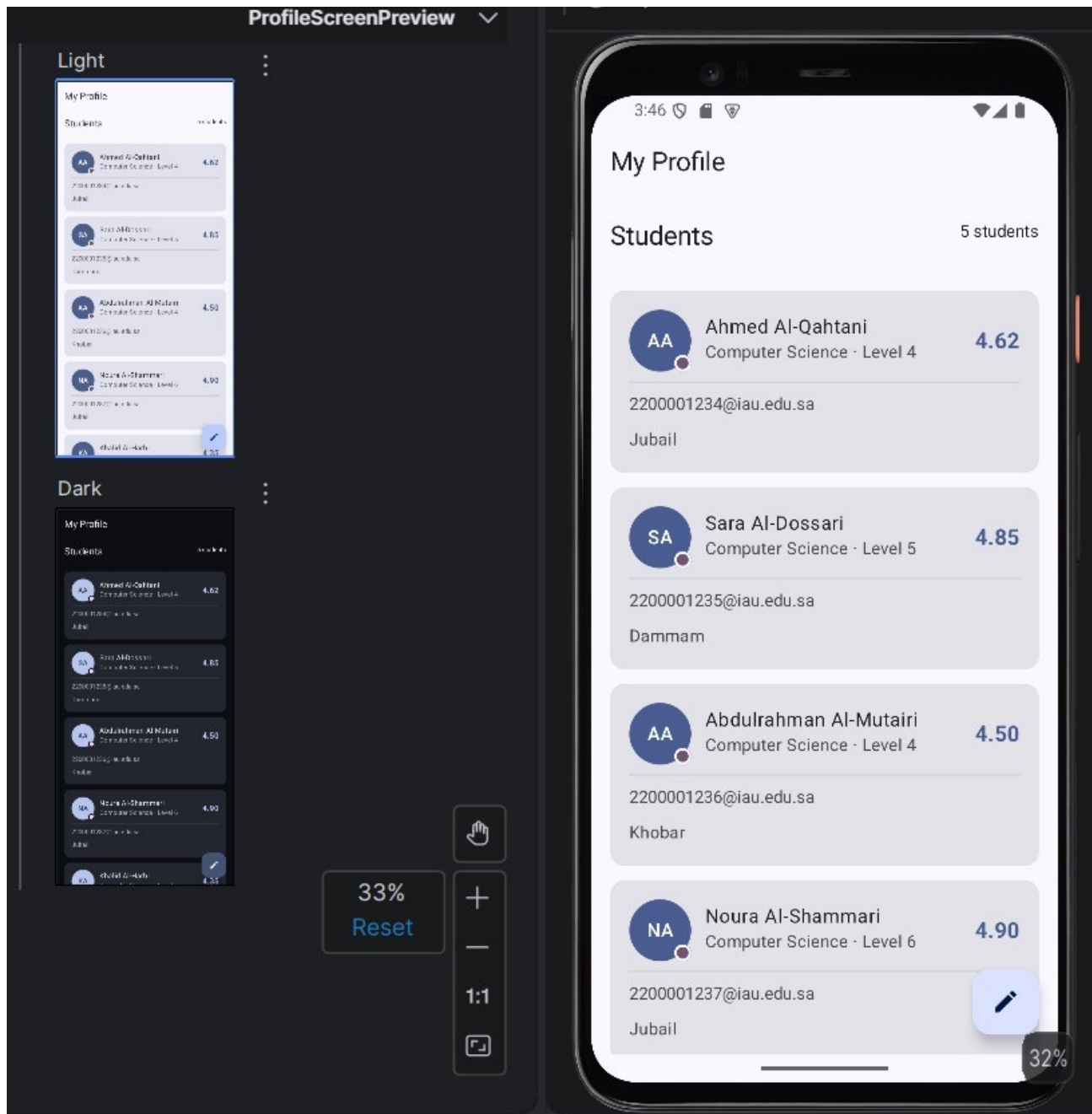
Profile Screen



Student List



Light and Dark List Previews



Student Card Code

```
package sa.edu.iau.cshj.lap3

import android.content.res.Configuration
import androidx.compose.foundation.background
import androidx.compose.foundation.border
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.CircleShape
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import sa.edu.iau.cshj.lap3.ui.theme.Lap3Theme

data class Student(
    val name: String,
    val program: String,
    val gpa: String,
    val email: String,
    val city: String
)

@Composable
fun StudentCard(student: Student, modifier: Modifier = Modifier) {
    Card(modifier = modifier.fillMaxWidth()) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Row(verticalAlignment = Alignment.CenterVertically) {
                Box(modifier = Modifier.size(52.dp)) {
                    Box(
                        modifier = Modifier.size(52.dp)
                            .clip(CircleShape)
                            .background(MaterialTheme.colorScheme.primary),
                        contentAlignment = Alignment.Center
                    ) {
                        Text(
                            text = student.name.trim().split(" ")
                                .filter { it.isNotEmpty() }.take(2)
                        )
                    }
                }
            }
        }
    }
}
```

Student Card Code Continued

```

        .joinToString("") { it.take(1).uppercase() },
        color = MaterialTheme.colorScheme.onPrimary,
        style = MaterialTheme.typography.titleMedium
    )
}
Box(
    modifier = Modifier.align(Alignment.BottomEnd)
    .size(14.dp).clip(CircleShape)
    .background(MaterialTheme.colorScheme.tertiary)
    .border(
        2.dp,
        MaterialTheme.colorScheme.surface,
        CircleShape
    )
)
}
Spacer(modifier = Modifier.width(12.dp))
Column(modifier = Modifier.weight(1f)) {
    Text(
        text = student.name,
        color = MaterialTheme.colorScheme.onSurface,
        style = MaterialTheme.typography.titleLarge
    )
    Text(
        text = student.program,
        color = MaterialTheme.colorScheme.onSurfaceVariant,
        style = MaterialTheme.typography.bodyMedium
    )
}
Text(
    text = student.gpa,
    color = MaterialTheme.colorScheme.primary,
    style = MaterialTheme.typography.titleLarge
)
}
HorizontalDivider()
Text(
    text = student.email,
    color = MaterialTheme.colorScheme.onSurfaceVariant,
    style = MaterialTheme.typography.bodyMedium
)
Text(

```

Student Card Code Continued

```

        text = student.city,
        color = MaterialTheme.colorScheme.onSurfaceVariant,
        style = MaterialTheme.typography.bodyMedium
    )
    }
}

@Preview(name = "Light", showBackground = true, widthDp = 400)
@Preview(
    name = "Dark", showBackground = true, widthDp = 400,
    uiMode = Configuration.UI_MODE_NIGHT_YES
)
@Composable
private fun StudentCardPreview() {
    Lap3Theme {
        StudentCard(
            student = Student(
                "Ahmed Al-Qahtani", "Computer Science · Level 4",
                "4.62", "2200001234@iau.edu.sa",
                "Jubail, Eastern Province"
            ),
            modifier = Modifier.padding(16.dp)
        )
    }
}

@Preview(name = "Long Name", showBackground = true, widthDp = 400)
@Composable
private fun LongNamePreview() {
    Lap3Theme {
        StudentCard(
            student = Student(
                "Abdulrahman Al-Mutairi", "Computer Science · Level 4",
                "4.62", "2200001234@iau.edu.sa",
                "Jubail, Eastern Province"
            ),
            modifier = Modifier.padding(16.dp)
        )
    }
}

```

Modifier Order Code

```
package sa.edu.iau.cshj.lap3

import androidx.compose.foundation.background
import androidx.compose.foundation.clickable
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.Text
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import sa.edu.iau.cshj.lap3.ui.theme.Lap3Theme

@Composable
fun ChainA(modifier: Modifier = Modifier) {
    Text(
        "content", color = Color.Black,
        modifier = modifier.background(Color(0xFF3DDC84))
            .padding(24.dp)
    )
}

@Composable
fun ChainB(modifier: Modifier = Modifier) {
    Text(
        "content", color = Color.Black,
        modifier = modifier.padding(24.dp)
            .background(Color(0xFF3DDC84))
    )
}

@Composable
fun ChainC(modifier: Modifier = Modifier) {
    Text(
        "content", color = Color.Black,
        modifier = modifier.clip(RoundedCornerShape(20.dp))
            .background(Color(0xFF3DDC84))
    )
}

@Composable
```

Modifier Order Code Continued

```

fun OrderPuzzle(modifier: Modifier = Modifier) {
    Column(
        modifier.padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text("A: background then padding")
        ChainA()
        Text("B: padding then background")
        ChainB()
        Text("C: clip then background")
        ChainC()
        TapTargetDemo()
    }
}

@Composable
fun TapTargetDemo(modifier: Modifier = Modifier) {
    var firstCount by remember { mutableStateOf(0) }
    var secondCount by remember { mutableStateOf(0) }
    Column(
        modifier,
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Clickable then padding: $firstCount",
            color = Color.Black,
            modifier = Modifier.background(Color(0xFFD7F5E4))
                .clickable { firstCount++ }.padding(16.dp)
        )
        Text(
            text = "Padding then clickable: $secondCount",
            color = Color.Black,
            modifier = Modifier.background(Color(0xFFD7F5E4))
                .padding(16.dp).clickable { secondCount++ }
        )
    }
}

@Preview(showBackground = true, widthDp = 400)
@Composable
private fun OrderPuzzlePreview() {
    Lap3Theme { OrderPuzzle() }
}

```

Profile Screen Code

```

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun ProfileScreen(modifier: Modifier = Modifier) {
    Scaffold(
        modifier = modifier.fillMaxSize(),
        topBar = { TopAppBar(title = { Text("My Profile") }) },
        floatingActionButton = {
            FloatingActionButton(onClick = {}) {
                Icon(
                    Icons.Default.Edit,
                    contentDescription = "Edit profile"
                )
            }
        }
    ) { innerPadding ->
        Column(
            modifier = Modifier.padding(innerPadding).fillMaxSize()
        ) {
            StudentCard(
                student = Student(
                    "Ahmed Al-Qahtani", "Computer Science · Level 4",
                    "4.62", "2200001234@iau.edu.sa",
                    "Jubail, Eastern Province"
                ),
                modifier = Modifier.padding(16.dp)
            )
        }
    }
}

```

Student List Code

```
package sa.edu.iau.cshj.lap3

import android.content.res.Configuration
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.material.icons.Icons
import androidx.compose.material.icons.filled.Edit
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import sa.edu.iau.cshj.lap3.ui.theme.Lap3Theme

private val students = listOf(
    Student(
        "Ahmed Al-Qahtani", "Computer Science · Level 4",
        "4.62", "2200001234@iau.edu.sa", "Jubail"
    ),
    Student(
        "Sara Al-Dossari", "Computer Science · Level 5",
        "4.85", "2200001235@iau.edu.sa", "Dammam"
    ),
    Student(
        "Abdulrahman Al-Mutairi", "Computer Science · Level 4",
        "4.50", "2200001236@iau.edu.sa", "Khobar"
    ),
    Student(
        "Noura Al-Shammari", "Computer Science · Level 6",
        "4.90", "2200001237@iau.edu.sa", "Jubail"
    ),
    Student(
        "Khalid Al-Harbi", "Computer Science · Level 5",
        "4.35", "2200001238@iau.edu.sa", "Dhahran"
    )
)

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun ProfileScreen(modifier: Modifier = Modifier) {
    Scaffold(
        modifier = modifier.fillMaxSize(),
        topBar = {
            TopAppBar(title = { Text("My Profile") })
        },
        floatingActionButton = {
            FloatingActionButton(onClick = {}) {
                Icon(
```

Student List Code Continued

```

        imageVector = Icons.Default.Edit,
        contentDescription = "Edit profile"
    )
}
}
) { innerPadding ->
    Column(
        modifier = Modifier.padding(innerPadding).fillMaxSize()
    ) {
        Row(
            modifier = Modifier.fillMaxWidth().padding(16.dp),
            horizontalArrangement = Arrangement.SpaceBetween
        ) {
            Text(
                "Students",
                style = MaterialTheme.typography.titleLarge
            )
            Text(
                "${students.size} students",
                style = MaterialTheme.typography.bodyMedium
            )
        }
        LazyColumn(
            modifier = Modifier.fillMaxWidth().weight(1f),
            contentPadding = PaddingValues(16.dp),
            verticalArrangement = Arrangement.spacedBy(12.dp)
        ) {
            items(students) { student ->
                StudentCard(student = student)
            }
        }
    }
}
}

@Preview(
    name = "Light", showBackground = true,
    widthDp = 400, heightDp = 850
)
@Preview(
    name = "Dark", showBackground = true,
    widthDp = 400, heightDp = 850,
    uiMode = Configuration.UI_MODE_NIGHT_YES
)
@Composable
private fun ProfileScreenPreview() {
    Lap3Theme { ProfileScreen() }
}

```

Main Activity Code

```
package sa.edu.iau.cshj.lap3

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import sa.edu.iau.cshj.lap3.ui.theme.Lap3Theme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Lap3Theme {
                ProfileScreen()
            }
        }
    }
}
```

Questions and Answers

Arrangement and Alignment

`verticalArrangement` distributes the children of a `Column` along its vertical main axis. `verticalAlignment` positions the children of a `Row` along its vertical cross axis, such as centring them vertically. Replacing the outer `Column` with a `Row` would arrange the header, divider, email and city horizontally, breaking the stacked card layout. The existing `verticalArrangement` argument would also be invalid for `Row`.

Modifier Order

The correct mapping is A to Result 2, B to Result 1 and C to Result 3. Composition establishes the UI and ordered modifier chain; layout measures and places the content, with padding changing the space available to the inner element. Drawing then paints the background and applies clipping at their positions in that chain, so background before padding covers the padded area, padding before background leaves an outer gap, and clip before background produces rounded corners. Placing clickable before padding includes the padded area in the tap target, while placing it after padding leaves that outer area outside the target.

Composable Functions and the Android Stack

UI composables such as `StudentCard` return `Unit` because they describe UI for Compose to manage rather than returning a widget object for the caller to mutate. When input or state changes, recomposition updates the UI description. If `Text` returned a mutable widget object, the caller would need to keep the reference and update its properties explicitly instead of declaring the text from state. This composable code belongs to the Applications layer of the Android software stack.

Dark Mode and Colour Roles

The supplied light and dark previews keep the card text readable because the card uses theme colours instead of fixed colours. `onSurface` provides the main text colour, `onSurfaceVariant` provides secondary text colours, and primary highlights the GPA. A fixed black name would have poor contrast on a dark card, while restoring `onSurface` would adapt the name to the active colour scheme.

Scaffold Content Padding

Ignoring `innerPadding` removes the offsets needed to keep content clear of the app bar and relevant system insets, allowing the card to overlap those areas. `Scaffold` passes these offsets as `PaddingValues` to its content lambda. The caller applies them to the appropriate container because `Scaffold` cannot know which nested layout should receive the padding or whether the content should intentionally draw behind its bars.

Modifier Results

Chain	Modifier Order	Correct Result
A	background then padding	Result 2
B	padding then background	Result 1
C	clip then background	Result 3